# PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

## 1. DATABASE ENTITY RELATIONSHIP ARCHITECTURE (ERD):

The ER diagram represents the core entities involved in the OptiCrop Smart Agricultural Production Optimization Engine and illustrates how they interact with each other. It provides a structured layout for managing data parameters cleanly.

## 2. ENTITIES AND PRIMARY KEYS MATRIX:

* User: uniquely identified by user_id

* SoilData: uniquely identified by soil_id

* Crop: uniquely identified by crop_id

* Dataset: uniquely identified by dataset_id

* MLModel: uniquely identified by model_id

* Prediction: uniquely identified by prediction_id

* Report: uniquely identified by report_id

## 3. STRUCTURAL RELATIONSHIPS & CARDINALITY:

* User to SoilData: One user can submit multiple soil data records for crop analysis (1 to Many).

* SoilData to Prediction: A single soil data entry can generate one crop prediction result (1 to One).

* Crop to Prediction: One crop can be recommended in multiple prediction results (1 to Many).

* MLModel to Prediction: One machine learning model can generate multiple prediction records (1 to Many).

* Dataset to MLModel: A single dataset can be used to train multiple machine learning models (1 to Many).

* Prediction to Report: One prediction can generate multiple agricultural reports and recommendations (1 to Many).

## 4. SYSTEM USE CASE COVERAGE:

This ER model supports the core functionalities of the OptiCrop system, including collecting soil metrics, processing models using machine learning estimators, and outputting smart analytical optimization recommendation views.